

1. ELECTROSTATICS (rot E = 0)

Resulting net force is 0 in a stable equilibrium: $\sum \vec{F}_{i \rightarrow j} = 0$

Coulomb's law: $\vec{F}_{1 \rightarrow 2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{\|\vec{x}_1 - \vec{x}_2\|^3} (\vec{x}_1 - \vec{x}_2)$ *unit vector*

$\vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{y})(\vec{x} - \vec{y})}{\|\vec{x} - \vec{y}\|^3} d^3y$ *conservative: $\vec{E}(\vec{x}) = -\vec{\nabla}\Phi(\vec{x})$*

Electric field: $\vec{\nabla} \times \vec{E} = -\vec{\nabla} \times \vec{\nabla}\Phi = 0$

$\vec{\nabla} \cdot \vec{E} = -\vec{\nabla} \cdot \vec{\nabla}\Phi = -\Delta\Phi = \frac{\rho}{\epsilon_0}$

Poisson equation: $\frac{1}{\epsilon_0} \rho(\vec{x}) = -\Delta\Phi(\vec{x})$ *no integral*

Charge density of a single charge at $\vec{y}$: $\rho(\vec{x}) = q\delta(\vec{x} - \vec{y})$ *Point charge in space: $\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{q}{\|\vec{x} - \vec{y}\|}$*

Gauss law: $\int_V d^3x \vec{\nabla} \cdot \vec{E}(\vec{x}) = \int_V d^3x \vec{\nabla} \cdot (\vec{\nabla}\Phi) = \int_V d^3x \Delta\Phi = \frac{1}{\epsilon_0} \int_V d^3x \rho(\vec{x})$

Stokes law: $\int_A d^2x \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \int_A d^2x \vec{\nabla} \times (-\Delta\Phi) = -\int_A d^2x \Delta\Phi$

Energy of an electric field is the work needed to generate a charge distribution

$W = \int d^3x \frac{1}{2} \epsilon_0 |\vec{E}(\vec{x})|^2$ *Energy density of E-field: $w = \frac{\epsilon_0}{2} |\vec{E}|^2$*

Self-energy: ∞ in total energy occur if we consider discrete distributions with $\rho(\vec{x}) = \sum q_i \delta(\vec{x} - \vec{x}_i)$

Heavisidefunktion: $\Theta(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$

$\int_{-\infty}^{\infty} f(x) \Theta(x-a) dx = \int_a^{\infty} f(x) dx$

$\int_{-\infty}^{\infty} f(x) \delta(x-a) dx = f(a)$

$\int_{-\infty}^{\infty} \delta(x) dx = 1$

$\int_{-\infty}^{\infty} f(x) \delta(x-a) dx = f(a)$

$\nabla^2 \frac{1}{\|\vec{x} - \vec{y}\|} = -4\pi\delta(\vec{x} - \vec{y})$

$\partial_x \delta(x-y) = -\partial_y \delta(x-y)$

Fourier: $\int_{-\infty}^{\infty} dt e^{-i\omega t} = 2\pi \delta(\omega)$

$\delta^3(\vec{x} - \vec{x}_0) = \frac{1}{r^2 \sin\theta} \delta(r-r_0) \delta(\theta-\theta_0) \delta(\varphi-\varphi_0)$

ex. sphere surface: $\rho(r) = \rho_0 \delta(r-R)$

surface charge distr.: $E_{\perp} = \frac{\sigma}{\epsilon_0}$ jump at $\frac{\sigma}{\epsilon_0}$, $E_{\parallel}$ stetig

Charged conductors: $C = \frac{Q}{V} = \frac{4\pi\epsilon_0 R}{1}$ (sphere)

$\vec{E} \cdot d\vec{s} = \vec{E} \cdot \vec{n} dA$ *Charge enclosed = $\sigma(\vec{x}) \Delta S$. $\vec{E}$ -field on surface: $\vec{E} = \frac{\sigma(\vec{x})}{\epsilon_0} \vec{n}$ & Energy density*

Force needed to unde deformation:

$dW = F \cdot dx = \int \frac{\partial W}{\partial x} dx$

Pressure on surface: $\frac{\partial W}{\partial V} = \frac{1}{3} \frac{\partial W}{\partial V} = \frac{1}{3} \frac{\partial}{\partial V} \int \frac{\epsilon_0}{2} |\vec{E}|^2 dV$

spherical coordinates: $r = \|\vec{x}\| = \sqrt{x^2 + y^2 + z^2}$

$\vec{x} = r \sin\theta \cos\varphi$, $y = r \sin\theta \sin\varphi$, $z = r \cos\theta$

$\nabla \cdot \vec{V} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 V_r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\sin\theta V_\theta) + \frac{1}{r \sin\theta \sin\varphi} \frac{\partial}{\partial \varphi} (\sin\theta \sin\varphi V_\varphi)$

Continuity Eq.: $\partial_t \rho + \vec{\nabla} \cdot \vec{j} = 0$

BOUNDARY CONDITION PROBLEMS

Green's functions: we want to solve: $-\nabla^2 \Phi(\vec{x}) = \frac{\rho(\vec{x})}{\epsilon_0}$

1st Id.: $\int_V d^3x (\phi \nabla^2 \psi - \psi \nabla^2 \phi) = \int_V d^3x (\vec{\nabla} \phi \cdot \vec{\nabla} \psi - \psi \nabla^2 \phi)$

2nd Id.: $\int_V d^3x (\phi \nabla^2 \psi - \psi \nabla^2 \phi) = \int_V d^3x (\vec{\nabla} \phi \cdot \vec{\nabla} \psi - \psi \nabla^2 \phi)$

Legendre polynomials: $P_n(\cos\theta)$

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spherical harmonics $Y_{lm}(\theta, \varphi)$

Multi pole expansion: $\Phi(\vec{x}) = \sum_{lm} a_{lm} Y_{lm}(\theta, \varphi)$

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2. MAGNETOSTATICS

Work & Energy: $P_{mech} = \int_V d^3x \vec{j} \cdot \vec{E}$

conservation of momentum: $\partial_t \vec{p} + \vec{\nabla} \cdot \vec{T} = 0$

Poynting's theorem (energy conservation): $\vec{\nabla} \cdot \vec{S} + \frac{\partial}{\partial t} \int_V d^3x u = 0$

Lorentz force: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

current density on disk: $\vec{j}(\vec{x}) = \vec{\omega} \times \vec{x}$

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Ampère: $\oint \vec{B} \cdot d\vec{s} = \mu_0 \int \vec{j} \cdot d\vec{A}$

Current density: $\vec{I} = \int \vec{j} \cdot d\vec{A}$

Spherical coord.: $\vec{j}(\vec{x}) = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \vec{j}_r)$

Faraday: change in magnetic flux $\Delta\Phi$ induces a current ΔI

Biot-Savart: current induces magn. field

Force: $\vec{F} = \int d^3x \vec{j} \times \vec{B}$

Dipole E-field: $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} (3(\vec{p} \cdot \hat{r})\hat{r} - \vec{p})$

Vector potential: $\vec{\nabla} \times \vec{A} = \vec{B}$

Ficktransf.: additional condition for potential $\vec{A}$

3. EM-STATICS IN MATTER

Dielectrics: $\vec{E} = \vec{E}_0 + \vec{E}_p$

Potential Energy: $W_{el}(\vec{x}) = \frac{1}{2} \int d^3x \rho(\vec{x}) \Phi(\vec{x})$

Capacitor: $C = \frac{Q}{V}$

Boundaries: $D_{\perp}$ stetig, $D_{\parallel}$ jump

4. MAXWELL EQUATIONS

Differential form: $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

Integral form: $\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$

Continuity Eq.: $\partial_t \rho + \vec{\nabla} \cdot \vec{j} = 0$

Induction: $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

5. CIRCUITS

$$\vec{\nabla} \times \vec{E} + \frac{1}{c} \frac{\partial \vec{A}}{\partial t} = 0 \rightarrow \mu_0 \epsilon_0 \vec{\nabla} \times \vec{E} = \mu_0 \epsilon_0 \frac{1}{c} \frac{\partial \vec{A}}{\partial t} = \vec{\nabla} \times \frac{1}{c} \frac{\partial \vec{A}}{\partial t} = \frac{1}{c} \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{A}) = \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = \Delta \vec{E} - \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} \xrightarrow{\text{TE}} \text{Telgr. eq. :}$$